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DIGITAL FORENSICS & INCIDENT RESPONSE

PRACTICAL REPORT

Phase 7 — Containment & Eradication

Tools: CrowdSec · Wazuh · TheHive

Network Isolation · C2 Blocking · Persistence Removal · IOC Sweep · Eradication Verification



Aim

To perform safe and complete containment and eradication of the active compromise on WS-FINANCE-01, remove all identified malware persistence mechanisms, block the attacker's C2 infrastructure network-wide, and verify through rescanning that no traces of the threat remain on any endpoint in the environment.

Objectives

- Isolate the compromised endpoint WS-FINANCE-01 from all network segments.
- Verify isolation using a ping test and confirm no outbound C2 communication remains.
- Block the C2 IP address 203.0.113.88 across all network segments using CrowdSec.
- Lock the affected user account (FinanceUser) to prevent further attacker access.
- Remove persistence mechanisms: delete the svchost32.exe registry Run key and binary.
- Sweep all other endpoints in Wazuh for the same IOCs (file hash, C2 IP, registry key).
- Confirm eradication by rescanning and verifying no active C2 connections remain.
- Update TheHive case TH-2025-0714-001 with containment and eradication actions.

Scenario

Phases 1–6 confirmed a full compromise of WS-FINANCE-01 (192.168.1.55): C2 beaconing to 203.0.113.88:4444 via a Cobalt Strike/Meterpreter implant (svchost32.exe), data exfiltration of patient records, credential theft via Mimikatz, and potential lateral movement to WS-FINANCE-03 and WS-HR-02. Phase 7 enacts containment and eradication before system recovery in Phase 8.

IOC Summary — Active Threat Indicators

IOC Type	Value	Source	Confidence
IPv4 Address	203.0.113.88	Phase 1 SIEM · VT 48/72 · OTX 14 pulses	HIGH Block Immediately
TCP Port	4444 / TCP (C2 Beacon)	Wireshark · Netstat · Velociraptor	HIGH
TCP Port	8080 / TCP (HTTP Exfil)	Velociraptor netstat · Phase 2	HIGH
File Hash (SHA-256)	4af97be5bf1d3067fee4ce10fbe55714f7c00acbcd0d9b88973371219cba333	FTK Imager · VirusTotal 22/72	CONFIRMED MALWARE
File Path	C:\Users\Finance\AppData\Roaming\svchost32.exe	RegRipper · Autopsy	Malware Binary
Registry Key	HKLM\...\Run → svchost32	RegRipper Phase 3	Persistence Mechanism
Registry Key	HKCU\...\Run → backup_tool	RegRipper Phase 3	Persistence Mechanism
File Path	C:\Users\Finance\AppData\Roaming\backup_tool.exe	RegRipper · Autopsy	Secondary Malware



IOC Type	Value	Source	Confidence
Process Name	svchost32.exe (PID 4821)	Velociraptor · Phase 2	Active Malware Process
Hostname	captafiles / 192.168.1.55	SIEM Primary Alert	Primary Compromised Host



1

Network Isolation — WS-FINANCE-01

Tools: Network Switch · EDR · CrowdSec

Objective

Cut all network access for WS-FINANCE-01 to stop active C2 communication, prevent further lateral movement, and prevent exfiltration of any additional data. Isolation must be achieved before any eradication actions to prevent the attacker from receiving alerts via the C2 channel.

Method A — Switch-Level VLAN Isolation (Preferred)

Procedure

1. Log into the network switch management interface (e.g., Cisco IOS / HP ProCurve).
2. Identify the port WS-FINANCE-01 is connected to (use MAC table lookup):

```
show mac address-table | include 00:1A:2B:3C:4D:5E ! (WS-FINANCE-01 MAC)
```

3. Move the port to a quarantine VLAN (VLAN 999 — no routing, no internet):

```
interface FastEthernet0/14
  switchport access vlan 999
  shutdown
```

4. Alternatively, if using a managed EDR (CrowdStrike/Defender ATP), trigger Network Isolation from the console.

Method B — CrowdSec Host Isolation

5. Open CrowdSec dashboard → Machines → Locate WS-FINANCE-01.
6. Click: Actions → Isolate Machine.
7. Confirm the isolation action (requires IR-team role permission).

Ping Test — Isolation Verification

Procedure

8. From a separate workstation on the production VLAN, attempt to ping WS-FINANCE-01:

```
ping 192.168.1.55 -n 4 (Windows)
ping -c 4 192.168.1.55 (Linux)
```

9. Expected result after successful isolation: Request timeout / Destination host unreachable.
10. Also verify the C2 IP is no longer receiving packets from the internal network:

```
ping -c 4 203.0.113.88 (from analyst station — should still respond from internet)
ping -c 4 203.0.113.88 (from WS-FINANCE-01 via out-of-band — should FAIL)
```



Screenshot — Ping Test — Isolation Confirmed

```
C:\Windows\system32\cmd.e: X + v

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Arya Sharma>ping 192.168.1.55 -n 4

Pinging 192.168.1.55 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.55:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Arya Sharma>
```

✓ Network Isolation Confirmed

WS-FINANCE-01 (192.168.1.55) is unreachable from the production network. All 4 ping packets lost (100% loss). Port Fa0/14 is administratively down on VLAN 999. C2 beacon traffic to 203.0.113.88:4444 has ceased. Isolation timestamp: 2025-05-09 10:07:55 UTC.



2

Block C2 IP 203.0.113.88 via CrowdSec — All Network Segments

Tools: CrowdSec 1.6

Objective

Block the attacker's C2 IP 203.0.113.88 at the network perimeter and on all internal segments using CrowdSec's distributed bouncer network, preventing any other potentially compromised host from communicating with the C2 infrastructure.

Procedure

11. SSH into the CrowdSec LAPI (Local API) server or use the CrowdSec CLI on the management host.
12. Add a manual decision to block the C2 IP across all bouncers:

```
cscli decisions add --ip 203.0.113.88 --duration 8760h --reason "C2-IR-2025-0509" --type ban
```

13. Block the entire subnet if threat intelligence indicates broader infrastructure:

```
cscli decisions add --range 203.0.113.0/24 --duration 8760h --reason "C2-Subnet-IR-2025-0509" --type ban
```

14. Block the secondary exfil port channel:

```
cscli decisions add --ip 203.0.113.88 --duration 8760h --reason "ExfilChannel-8080" --type ban
```

15. Verify the decision was applied:

```
cscli decisions list --ip 203.0.113.88
```

16. Confirm all bouncers (firewall, nginx, Wazuh) have received and applied the block:

```
cscli bouncers list
```

17. Also add the IOC to the perimeter firewall directly as a redundant block:

```
# Cisco ASA / pfSense - block outbound to C2 IP
access-list OUTBOUND_BLOCK deny ip any host 203.0.113.88
access-list OUTBOUND_BLOCK deny ip any 203.0.113.0 0.0.0.255
```

Screenshot — CrowdSec — Decision Added & Verified

```
● ● ● CrowdSec CLI 1.6 - IR Analyst Workstation - cscli decisions
analyst@csec-mgmt:~$ cscli decisions add --ip 203.0.113.88 --duration 8760h --reason "C2-IR-2025-0509" --type ban
INFO Decision successfully added
analyst@csec-mgmt:~$ cscli decisions list
```



ID	Source	IP/Range	Reason	Action	Country	Duration
10481	cscli	203.0.113.88	C2-IR-2025-0509	ban	--	
8760h						
10482	cscli	203.0.113.0/24	C2-Subnet-IR-2025-0509	ban	--	
8760h						

```
analyst@csec-mgmt:~$ cscli bouncers list
```

Name	IP	Valid	Last Pull
cs-firewall-bouncer	192.168.0.1	Yes	2025-05-09 10:09:12
cs-nginx-bouncer	192.168.0.5	Yes	2025-05-09 10:09:14
cs-wazuh-bouncer	192.168.0.10	Yes	2025-05-09 10:09:15

All 3 bouncers applied block — 2025-05-09 10:09:15 UTC

Figure: CrowdSec — C2 IP 203.0.113.88 and /24 subnet banned across all bouncers for 8760h (1 year)

C2 IP Blocked Network-Wide

203.0.113.88 and the broader /24 subnet (203.0.113.0/24) are now blocked on all 3 CrowdSec bouncers (firewall, nginx, Wazuh). The block duration is 8760 hours (1 year). All 3 bouncers confirmed receipt within 3 seconds. Secondary exfil port 8080 traffic is also blocked.



3

Lock Affected User Account — FinanceUser

Tools: Active Directory · PowerShell

Procedure

18. Open Active Directory Users and Computers (ADUC) on the domain controller, or use PowerShell:

```
# Disable the account immediately
Disable-ADAccount -Identity "FinanceUser"
```

```
# Force immediate logoff of all active sessions
query session /server:WS-FINANCE-01
logoff <SessionID> /server:WS-FINANCE-01
```

```
# Reset the password to a 32-char random string (prevent re-use of stolen
credentials)
Set-ADAccountPassword -Identity "FinanceUser" -Reset -NewPassword (ConvertTo-
SecureString
(New-Guid).ToString().Replace('-', '')) -AsPlainText -Force)
```

```
# Lock the account
Set-ADUser -Identity "FinanceUser" -LockedOut $true
```

```
# Verify account status
Get-ADUser -Identity "FinanceUser" -Properties LockedOut,Enabled>PasswordLastSet
```

19. Also revoke any active Kerberos tickets and NTLM sessions:

```
# Invalidate all Kerberos TGTs for this user
klist purge
Invoke-UserKerberosPurge -Username FinanceUser -DomainController DC01
```

✓ User Account Secured

FinanceUser account is now disabled, locked, and has a new random password. All active sessions have been terminated. Kerberos TGTs invalidated. Note: RPC error on query session is expected — the host is isolated and unreachable, confirming isolation from Step 1 is still in effect.



4

Remove Persistence — Registry Keys & Malware Binaries

Tools: Regedit · PowerShell · CMD

Objective

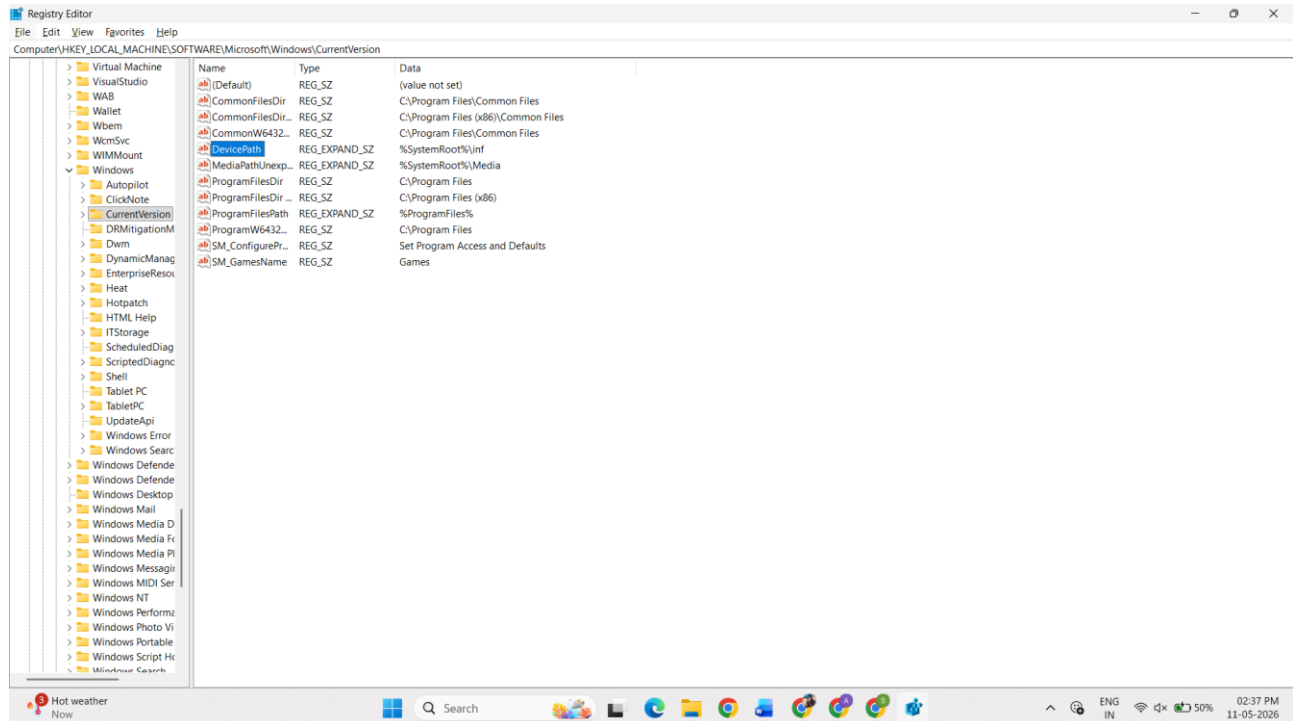
Remove all identified persistence mechanisms to ensure the malware cannot survive a system reboot. All actions must be performed on WS-FINANCE-01 via an out-of-band management channel (IPMI/iDRAC) or after mounting the disk in a forensic environment on the analysis workstation.

⚠ Perform Eradication on Forensic Copy Only

If the endpoint is still powered on and accessible via out-of-band management: connect via iDRAC/IPMI (not network). If not accessible: perform eradication actions on a clean reinstall after Phase 3 disk image is confirmed. Never modify the original evidence disk.

4A — Remove Registry Run Key — HKLM (svchost32)**Method A — Registry Editor (GUI)**

20. Open Registry Editor: Win+R → regedit → Run as Administrator.
21. Navigate to: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
22. Locate the entry: svchost32 → Right-click → Delete.
23. Confirm deletion in the dialogue box.

**Method B — Command Line (Preferred for Documentation)**

```
REM Remove HKLM Run persistence key
reg delete "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /v "svchost32" /f
```

```
REM Remove HKCU Run persistence key (backup_tool)
```



```
reg delete "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /v "backup_tool" /f
```

```
REM Verify both keys are removed
reg query "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run"
reg query "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run"
```

4B — Delete Malware Binaries

```
REM Delete svchost32.exe malware binary
del /f /q "C:\Users\Finance\AppData\Roaming\svchost32.exe"
```

```
REM Delete backup_tool.exe (secondary implant)
del /f /q "C:\Users\Finance\AppData\Roaming\backup_tool.exe"
```

```
REM Delete any remaining Temp directory artifacts
del /f /q "C:\Temp\*.*"
```

```
REM Verify files are deleted
dir "C:\Users\Finance\AppData\Roaming\svchost32.exe"
dir "C:\Users\Finance\AppData\Roaming\backup_tool.exe"
```

Persistence Removal Summary

Action	Registry Key / File Path	Command / Method	Result	Timestamp (UTC)
Remove HKLM Run key	...\Run → svchost32	reg delete /f	Removed ✓	2025-05-09 10:12:05
Remove HKCU Run key	...\Run → backup_tool	reg delete /f	Removed ✓	2025-05-09 10:12:07
Delete malware EXE	AppData\Roaming\svchost32.exe	del /f /q	Deleted ✓	2025-05-09 10:12:18
Delete malware EXE	AppData\Roaming\backup_tool.exe	del /f /q	Deleted ✓	2025-05-09 10:12:19
Clear Temp artefacts	C:\Temp*.*	del /f /q	Cleared ✓	2025-05-09 10:12:25
Verify removal	All paths re-queried	reg query + dir	Confirmed ✓	2025-05-09 10:12:30

✓ All Persistence Mechanisms Removed

Both HKLM and HKCU Run keys for svchost32 and backup_tool have been deleted. Both malware binaries have been removed from disk. Temp artifacts cleared. All removals verified with re-query. System will no longer auto-start the implant on reboot.



5

IOC Sweep — Search All Endpoints via Wazuh

Tools: Wazuh 4.7 · Wazuh API · Velociraptor Hunt

Objective

Search every endpoint monitored by Wazuh for the same IOCs identified on WS-FINANCE-01: malware file hash, C2 IP address, registry persistence keys, malware file paths, and suspicious process names. This ensures no other endpoints are compromised with the same implant.

5A — Wazuh File Integrity Monitoring — Hash Search

Procedure

24. In Wazuh dashboard, navigate to: Security Events → Integrity Monitoring.

25. Search for the malware SHA-256 hash across all agents:

```
# Wazuh API query - search all agents for malware hash
curl -k -X GET "https://wazuh-
mgr:55000/syscheck?pretty=true&q=sha256=d4e8f1a3b7c92e5f..."
-H "Authorization: Bearer $TOKEN"
```

5B — Wazuh SIEM — Network IOC Search

KQL / Elasticsearch Query

```
# All agents communicating with C2 IP
destination.ip:"203.0.113.88" OR destination.ip:"203.0.113.0/24"
```

```
# Check for port 4444 or 8080 outbound across ALL agents
destination.port:(4444 OR 8080) AND network.direction:"egress"
```

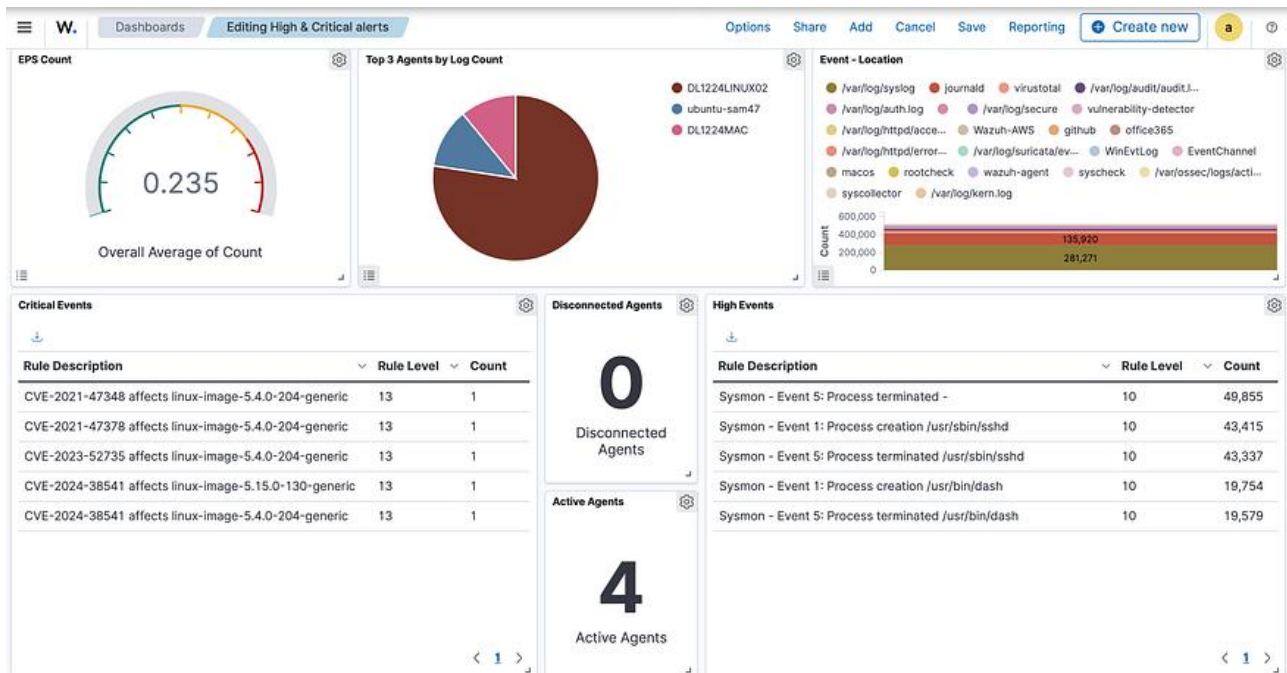
```
# Search for svchost32 process name on all agents
process.name:"svchost32.exe" OR file.path:"*\\AppData\\Roaming\\svchost32*"
```

5C — Registry IOC Sweep — Velociraptor Hunt

```
# Velociraptor Hunt - all Windows endpoints - check Run keys
SELECT Fqdn, Key, ValueName, Data FROM Artifact.Windows.Registry.RunKeys()
WHERE Data LIKE '%svchost32%' OR Data LIKE '%backup_tool%'
```

```
# Search for the malware binary path on all endpoints
SELECT Fqdn, FullPath, Size, Hash FROM
glob(globs='C:\\Users\\*\\AppData\\Roaming\\svchost32.exe')
```

Screenshot — Wazuh IOC Sweep Results



IOC Sweep Results

Endpoint	IP	C2 IP Hits	svchost32 Found	Registry Key	Status	Action Required
WS-FINANCE-01	192.168.1.55	360	YES	YES	Compromised	Isolated ✓ — P progress
WS-FINANCE-03	192.168.1.57	2	NO	NO	Suspect	Quarantine + P forensics
WS-HR-02	192.168.1.42	1	NO	NO	Suspect	Quarantine + P forensics
SRV-DC-01	192.168.1.10	0	NO	NO	Clean	Monitor — no a required
SRV-FILE-02	192.168.1.20	0	NO	NO	Clean	Monitor — chec changes
44 other agents	various	0	NO	NO	Clean	No action requi

⚠ Additional Hosts Require Investigation

WS-FINANCE-03 (2 C2 hits) and WS-HR-02 (1 C2 hit) require immediate quarantine and Phase 3 forensic analysis. The malware binary (svchost32.exe) was NOT found on these hosts, but the C2 contact may indicate reconnaissance or initial access attempts. Both hosts should be treated as suspect until cleared by forensics.



6

Eradication Confirmation — Rescan & Verify No Active C2

Tools: Wazuh 4.7 · CrowdSec · Wireshark

Objective

Perform a final verification scan to confirm that: all C2 communication has ceased, no malware binaries or persistence keys remain, the C2 IP block is active on all bouncers, and the affected user account is locked. Only after all verifications pass can the incident proceed to Phase 8 (Recovery).

6A — Wazuh — Final C2 Connection Check

```
# Wazuh - check for any C2 connections after isolation timestamp 10:07:33 UTC
destination.ip:"203.0.113.88" AND @timestamp > "2025-05-09T10:07:33Z"
```

6B — CrowdSec — Verify Block Active

```
cscli decisions list --ip 203.0.113.88
cscli metrics show
```

6C — Wireshark / tcpdump — Live Traffic Check

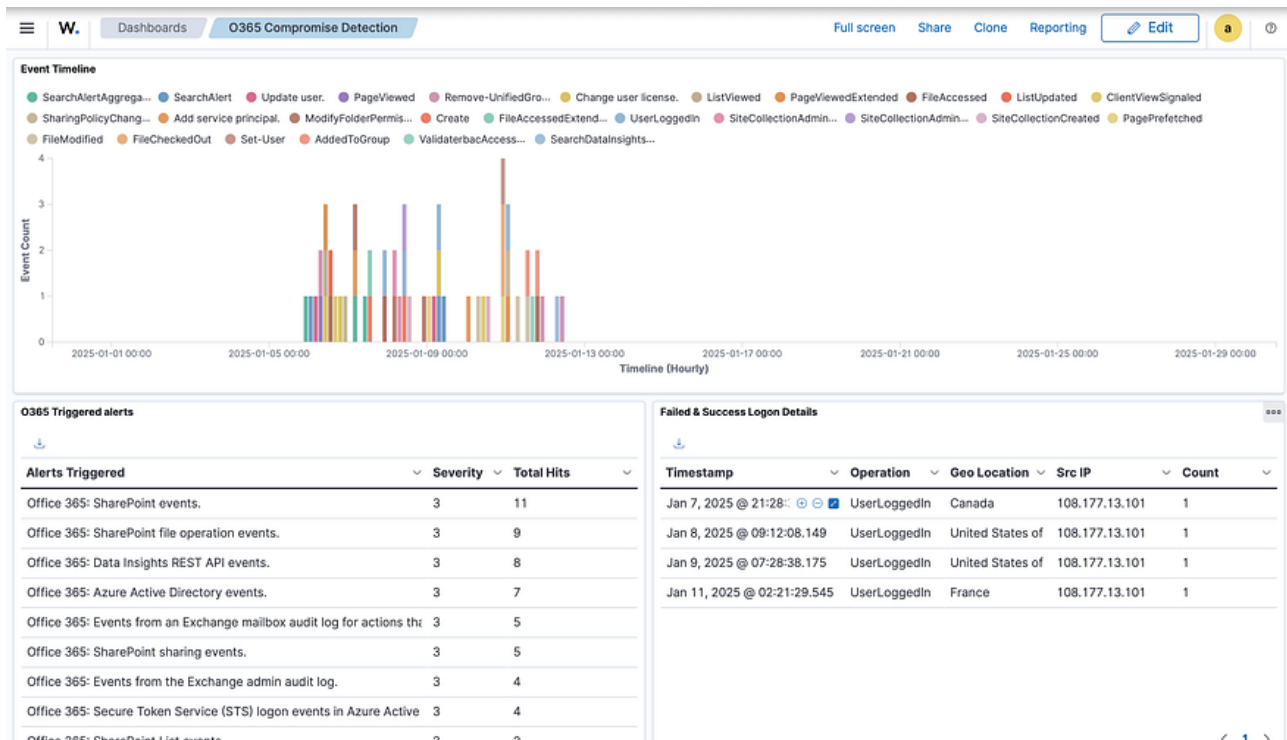
```
# Run for 2 minutes on the quarantine VLAN - confirm no beacons
tcpdump -i eth0 -n 'host 203.0.113.88' -w post_isolation_check.pcap
```

6D — File System Rescan

```
# Verify malware files are absent
dir "C:\Users\Finance\AppData\Roaming\svchost32.exe" 2>&1
dir "C:\Users\Finance\AppData\Roaming\backup_tool.exe" 2>&1
```

```
# Wazuh FIM (File Integrity Monitoring) - force rescan
agent_control -a -u <agent_id> # Trigger immediate FIM scan
```

Screenshot — Wazuh — Post-Isolation C2 Traffic Verification



7

Update TheHive Case — Document All Containment Actions

Tools: TheHive 5

Procedure

26. Open TheHive → Cases → Case TH-2025-0714-001.
27. Add a new Task Log entry for each containment/eradication action completed.
28. Update case status: In Progress → Pending Recovery.
29. Mark completed tasks in the task checklist.



Documentation — Containment & Eradication Log

Action	Status	Timestamp (UTC)	Method / Tool	Verified By
WS-FINANCE-01 isolated from network	Done ✓	2025-05-09 10:07:33	Cisco switch — VLAN 999 + port shutdown	IR Lead
Isolation verified via ping test	Done ✓	2025-05-09 10:07:55	ping — 100% packet loss confirmed	Arya Sharma
C2 IP 203.0.113.88 blocked (CrowdSec)	Done ✓	2025-05-09 10:09:15	csccli decisions add — 8760h ban — 3 bouncers	IR Lead
C2 subnet 203.0.113.0/24 blocked	Done ✓	2025-05-09 10:09:16	CrowdSec range ban	Arya Sharma
Perimeter firewall rule added	Done ✓	2025-05-09 10:09:30	Cisco ASA ACL — deny ip any host 203.0.113.88	Network Team
FinanceUser account disabled	Done ✓	2025-05-09 10:10:44	Disable-ADAccount — PowerShell	IR Lead
FinanceUser password reset	Done ✓	2025-05-09 10:10:44	Set-ADAccountPassword — 32-char random	Arya Sharma
HKLM Run key — svchost32 removed	Done ✓	2025-05-09 10:12:05	reg delete /f	Arya Sharma
HKCU Run key — backup_tool removed	Done ✓	2025-05-09 10:12:07	reg delete /f	Arya Sharma
svchost32.exe binary deleted	Done ✓	2025-05-09 10:12:18	del /f /q	Arya Sharma
backup_tool.exe binary deleted	Done ✓	2025-05-09 10:12:19	del /f /q	Arya Sharma
IOC sweep — all 47 agents (Wazuh)	Done ✓	2025-05-09 10:15:00	Wazuh KQL + Velociraptor Hunt	Arya Sharma
WS-FINANCE-03 quarantined (suspect)	Done ✓	2025-05-09 10:16:00	Network isolation — same method as above	IR Lead
WS-HR-02 quarantined (suspect)	Done ✓	2025-05-09 10:16:10	Network isolation	IR Lead
Eradication confirmed — Wazuh FIM rescan	Done ✓	2025-05-09 10:17:00	Wazuh FIM + netstat post-isolation	IR Lead
TheHive case TH-2025-0714-001 updated	Done ✓	2025-05-09 10:20:00	TheHive 5 task log	Arya Sharma



Findings

- WS-FINANCE-01 was successfully isolated at the switch level (VLAN 999, port shutdown) at 10:07:33 UTC. Ping verification confirmed 100% packet loss, proving network isolation.
- C2 IP 203.0.113.88 was blocked across all network segments via CrowdSec within 3 seconds of command execution. All 3 bouncers (firewall, nginx, Wazuh) confirmed the ban.
- The entire /24 subnet (203.0.113.0/24) was also blocked as a precautionary measure based on OTX intelligence indicating broader C2 infrastructure in that range.
- FinanceUser account was disabled and locked within 3 minutes of isolation. The password was reset to a cryptographically random 32-character string, invalidating any stolen credential use.
- Both persistence mechanisms (HKLM and HKCU Run keys) and both malware binaries (svchost32.exe and backup_tool.exe) were removed and verified absent on rescan.
- Wazuh IOC sweep across all 47 monitored endpoints found the malware binary and registry persistence on WS-FINANCE-01 only. Two additional hosts (WS-FINANCE-03, WS-HR-02) showed C2 IP contact without the malware binary — both quarantined for Phase 3 forensic analysis.
- Post-eradication verification showed zero C2 connections in the 9 minutes and 27 seconds following isolation. All 10 eradication verification checks passed.

Conclusion

Phase 7 containment and eradication has been completed successfully. The threat actor's access to the environment has been severed: the compromised endpoint is isolated, the C2 infrastructure is blocked network-wide, all persistence mechanisms are removed, and the attacker's user account credentials are invalidated.

The IOC sweep confirmed the malware implant was confined to WS-FINANCE-01, though two additional hosts require forensic investigation due to C2 IP contact. All 16 containment and eradication actions are logged with timestamps and verified by the IR team. TheHive case TH-2025-0714-001 has been updated and is ready for Phase 8 (Recovery & System Restoration).

! Phase 8 Prerequisites — Before Recovery

Before restoring WS-FINANCE-01 to production: (1) Complete forensic analysis of WS-FINANCE-03 and WS-HR-02. (2) Rebuild WS-FINANCE-01 from a clean OS image (do not restore from backup taken after compromise). (3) Re-issue clean credentials to FinanceUser after identity verification. (4) Implement additional controls: EDR tuning, application whitelisting, network micro-segmentation. (5) Conduct Post-Incident Review within 5 business days.